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Li^+ solvation and kinetics of Li^+–BF_4^-/PF_6^- ion pairs in ethylene carbonate. A molecular dynamics study with classical rate theories

Tsun-Mei Chang[1] and Liem X. Dang[2]
[1]*Department of Chemistry, University of Wisconsin–Parkside, Kenosha, Wisconsin 53141, USA*
[2]*Physical Sciences Division, Pacific Northwest National Laboratory, Richland, Washington 93352, USA*

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Using our polarizable force-field models and employing classical rate theories of chemical reactions, we examine the ethylene carbonate (EC) exchange process between the first and second solvation shells around Li^+ and the dissociation kinetics of ion pairs Li^+–$[BF_4]$ and Li^+–$[PF_6]$ in this solvent. We calculate the exchange rates using transition state theory and correct them with transmission coefficients computed by the reactive flux, Impey, Madden, and McDonald approaches, and Grote-Hynes theory. We found that the residence times of EC around Li^+ ions varied from 60 to 450 ps, depending on the correction method used. We found that the relaxation times changed significantly from Li^+–$[BF_4]$ to Li^+–$[PF_6]$ ion pairs in EC. Our results also show that, in addition to affecting the free energy of dissociation in EC, the anion type also significantly influences the dissociation kinetics of ion pairing. *Published by AIP Publishing.* [http://dx.doi.org/10.1063/1.4991565]

I. INTRODUCTION

In recent years, much effort has focused on developing new energy storage systems and portable power supplies for alternative energy sources and consumer electronics. Because of their high-energy-storage density and high output voltage, rechargeable lithium-ion batteries (LIBs) are one of the most versatile battery systems yet developed, and they are used extensively in electronic devices including laptop computers and mobile phones and also to store energy needed to power electric vehicles. These LIBs play a significant role in energy storage applications. The capacity of LIBs depends on the electrode materials used, and the energy output is determined by the chemical compositions of the electrodes. To understand this effect, computer simulations and statistical mechanical methods are used to provide a molecular-level understanding of LIB properties and for designing new materials for use in LIBs. Many studies have been carried out to investigate the effects of various types of electrode materials on battery performance.[1–5]

The performance of these LIBs is also impacted by the other major component of the battery system—the electrolyte being used. Ideally, the electrolyte should be very stable and should enhance Li^+-ion mobility. Aprotic solvents such as ethylene carbonate (EC) and propylene carbonate (PC) are widely used as liquid electrolytes of these battery systems.[6,7] Experimental and theoretical studies have been performed to understand Li^+-ion transport and solvation in various carbonate electrolytes. Previous studies have shown that the Li^+ ion prefers to interact with carbonyl oxygen atoms of the carbonate molecule and that the Li^+-ion has a coordination number of 4. In mixed carbonate systems such as EC/PC, both EC and PC participate in solvating the Li^+ ion.[8–16]

Computer simulations have been shown to provide molecular understanding of the chemical processes; however, the reliability of the simulated results critically depends on the accuracy of the interaction potential models. From the computational perspective, classical pair-wise potentials sometimes are limited in their transferability and their ability to describe polarization effects. In this work, we used experimental data to construct a polarizable potential model for EC and then explored the consequences of polarizable force fields by examining the structural and thermodynamic properties of Li^+ ions and lithium salts in liquid EC.

Our approach was to apply rate theory methods to study solvation of Li^+ ions and Li^+-ion pairings in LIBs. Here, we report the results from computer simulations of the exchange dynamics around solvated Li^+ ions in liquid EC. Additionally, we provide details of Li^+–$[BF_4]$/$[PF_6]$ ion pairing kinetics in EC to examine the anion effects of lithium salts. Using Fourier-transform infrared experiments, Tominaga and co-workers examined the anion effect on Li^+-ion conductivity.[17] By varying the anion composition of the lithium salts, they found that the anion played an important role in determining the lithium transference numbers and ionic conductivities. Aravindan and co-workers also studied the effects of anions of the lithium salts on the ionic mobility and their solid–electrolyte interphase formation towards carbonaceous anodes for lithium rechargeable batteries.[18] Our results show that in addition to affecting the free energy of ion pair dissociation in EC, the anion also considerably influences the kinetics of ion pairings. These results will increase our understanding of the dynamic and kinetic properties of LIB systems. The remainder of this paper is organized as follows. In Sec. II, we describe the potential models and simulation methods used. Results and discussion are presented in Sec. III, and our conclusions are in Sec. IV.

II. POTENTIAL MODEL AND COMPUTATIONAL METHODS

In this study, we use polarizable potential models to describe the interactions between all ionic and molecular species. The force-field parameters for Li^+, $[BF_4]$, and $[PF_6]$

TABLE I. Potential parameters used in this study.

Atom type	σ(Å)	ε(kcal/mol)	q(e)	α(Å^3)
C	3.75	0.105	0.7867	1.2758
C3	3.40	0.115	0.2374	0.8896
O	2.98	0.210	−0.4873	0.6061
OS	3.00	0.170	−0.3921	0.6005
H	2.51	0.0285	0.0025	0.4516

were previously developed by our group.[19] We have extended our work to the development of a polarizable model that describes the interaction of EC. The potential model is based on an all-atom framework and includes the effect of induced polarization. Each atom is assigned a partial charge, Lennard-Jones parameters, and a point polarizability. The partial charges are taken from *ab initio* calculations based on electrostatic surface potential fitting and further adjusted to reproduce the gas-phase dipole moment of 4.9 D.[20] Most previous potential models of EC use a gas-phase dipole moment of ∼6 D to account for the induced polarization effect in bulk. Atomic polarizabilities are taken from the work of Duan and workers.[21] The Lennard-Jones parameters are optimized by carrying out a series of classical molecular dynamics (MD) simulations of bulk liquid EC at 313 and 423 K. During the parameter optimizations, we calculated the thermodynamic properties of EC including the enthalpy of vaporization and densities and compared our calculated values to values obtained from experiments. The final potential parameters are obtained when the results from the potential model adequately reproduce the experimental values (see Table I).

MD simulations were performed on bulk liquids at several temperatures. All systems contained 1728 molecules in a simulation cell of linear dimensions approximately equal to 60 × 60 × 60 Å. We applied periodic boundary conditions in all three spatial directions. The simulations were carried out in a constant pressure-temperature ensemble using the Langevin dynamics thermostat. We employed the particle mesh Ewald method to handle the long-range Coulombic interactions[22] and used a time step of 2 fs to integrate the equations of motion. All simulations were carried out using a modified version of the Amber 9 software package.[23] To study the solvent exchange and kinetics of Li^+, $[BF_4]$, and $[PF_6]$ in EC, simulations were carried out with one Li^+ ion or one Li^+–$[BF_4]$/$[PF_6]$ ion pair in 800 EC molecules in a cubic box with a linear dimension of ∼44 Å under constant volume-temperature conditions for potential of mean force (PMF) calculations and constant pressure-temperature conditions for structural investigations. We maintained the temperature at 330 K and used the SHAKE algorithm to fix bond lengths that involves H atoms.[24] We equilibrated all systems for at least 2 ns prior to a 10 ns simulation period during which data were collected.

III. RESULTS AND DISCUSSIONS

A. Bulk ethylene carbonate

From a series of MD simulations, we optimized the final potential parameters of EC by fitting the calculated liquid densities and enthalpies of vaporization to the experimental data at 313 and 423 K. During simulations, the enthalpy of vaporization can be evaluated by using the following relation:

TABLE II. Computed and experimental properties of liquid EC.[26]

Temperature (K)	Density (g/cm^3) MD (exp)	ΔH_{vap} (kcal/mol) MD (exp)	Diffusion constant (10^{-10} m^2/s)
313	1.32 (1.32)	15.4	0.98
330	1.31	15.1	1.4
380	1.26	14.3 (14.5[a])	3.2
423	1.22 (1.21)	13.6 (13.5)	5.6

[a]This value is estimated from the vapor pressure data of ethylene carbonate from 368 to 449 K by Hong and co-workers.[35]

$$\Delta H_{vap} = E_{vap} - E_{liq} + RT, \tag{1}$$

where E_{vap} and E_{liq} are the potential energies in the vapor and liquid phase, respectively, R is the ideal gas constant, and T is the absolute temperature in K. The computed and experimental properties are listed in Table II. We observed satisfactory agreement between the simulation results and the experimental results, which suggest that our potential model is able to capture the thermodynamic properties of bulk EC. Over the entire temperature range, discrepancies between the experimental and simulated liquid densities are less than 1%.[25]

To understand the transport properties of EC, we examined the translational dynamics of EC using mean-square displacements (MSD), $\langle \Delta r^2(t)\rangle$. From MSD, the long-time diffusive behavior can be described using the Einstein relation, and the self-diffusion constant, D, of EC can be estimated by

$$D = \lim_{t\to\infty} \frac{1}{6t} \left\langle |\vec{r}(t) - \vec{r}(0)|^2 \right\rangle = \lim_{t\to\infty} \frac{1}{6t} \left\langle \Delta r^2(t) \right\rangle, \tag{2}$$

where $\vec{r}(t)$ is the center-of-mass coordinate of the EC at time t and $\langle \ldots \rangle$ denotes an ensemble average. Figure 1 shows $\langle \Delta r^2(t)\rangle$ as a function of time for EC from MD simulations at 313 K, 330 K, 380 K, and 423 K. It can be observed that

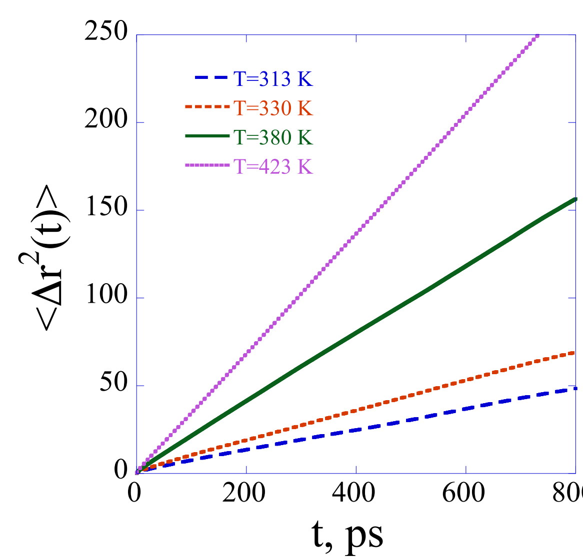

FIG. 1. Computed mean-square displacement of EC as a function of time at 313, 330, 380, and 423 K.

the MSD follows linear diffusive motion at longer time periods, and the diffusion rates are then extracted from MSD using least-square fitting (see Table II). The diffusion constants of EC are estimated to be 0.98×10^{-10} and 5.6×10^{-10} m^2/s in this temperature range. These values are a factor of two lower than the experimental values reported by Hayamizu.[26]

To understand the liquid structures of bulk EC, the radial distribution functions (RDFs) of the EC at 330 K are examined. Figure 2 shows the RDFs between O–O, O–OS, O–C3, and O–H atom pairs (see Table I for the atomic labels) at 330 K as a function of atomic separation. Distinct features are observed in these RDFs at interatomic distances up to 16 Å. The first peak positions of the O–O, O–OS, O–C3, and O–H atomic RDFs are found to be at 3.85 Å, 4.55 Å, 3.48 Å, and 2.75 Å, respectively. These results indicate that, as expected, the carbonyl O prefers to associate with H atoms. The computed RDFs have features that are similar to those reported by Freitas and co-workers based on Monte Carlo simulations.[27]

We further characterize the liquid structure by examining the static structure function that can be evaluated based on the procedure outlined in recent work using Debye's scattering equation,

$$I(Q) = \sum_i \sum_j f_i f_j x_i x_j \frac{\sin(Qr_{ij})}{Qr_{ij}}, \tag{3}$$

where Q is the wave vector and i and j refer to different atoms with the atomic scattering factors f_i and f_j, respectively. The term x_i represents the number density of atom type i and r_{ij} is the distance between atoms i and j. The scattering function is then multiplied by a factor of Q and a Q-dependent sharpening factor, M(Q), as shown in the following relationship:

$$M(Q) = \frac{f_O^2(0)}{f_O^2(Q)} \exp(-0.01Q^2), \tag{4}$$

where f_O is the atomic scattering factor for the oxygen atom. The calculated function QI(Q)M(Q) of EC at 330 K is plotted as a function of the wave vector Q in Fig. 3. Peaks are observed at Q = 3.8, 6.1, 9.3, and 11.5 $Å^{-1}$ in the structure function.

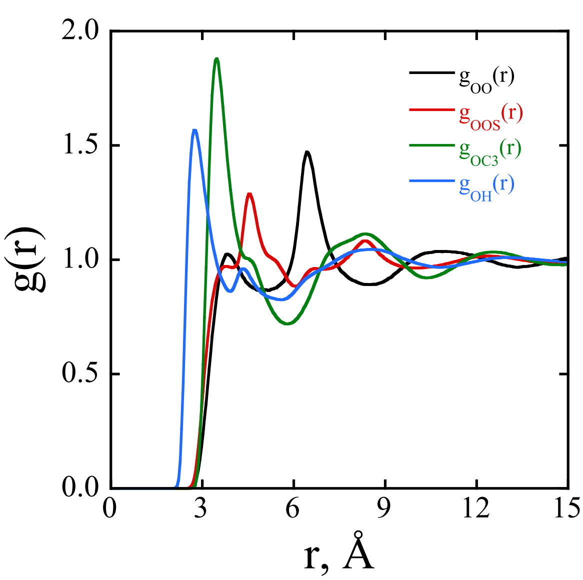

FIG. 2. Atomic radial distribution functions of EC at 330 K.

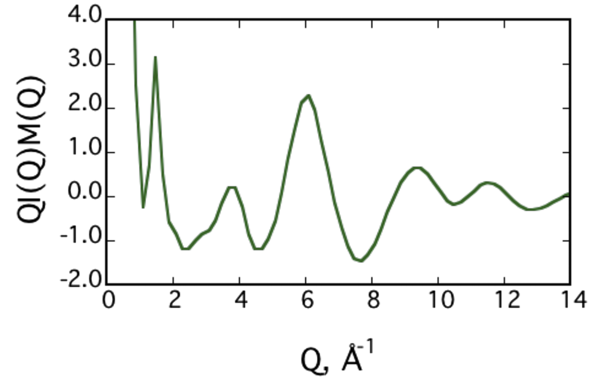

FIG. 3. Calculated structure factor QI(Q)M(Q) for EC at 330 K.

These results can be directly compared with experimental data obtained from X-ray diffraction measurements performed by Soetens.[28] In general, the simulations can reproduce the overall features of the experimental structure function except at small wave vectors, and the peak positions of the simulated structure factor are in excellent agreement with the corresponding experiment.

B. Properties of Li^+ ions in liquid EC

To examine the structure of Li^+ ions in EC, Li^+-solvent RDFs are determined from a single Li^+ ion in 800 EC molecules. The RDFs for Li^+–O, Li^+–C, Li^+–OS, and Li^+–C3 atom pairs are shown in Fig. 4. The sharp first peak in $g_{LiO}(r)$ (with a magnitude of about 44) positioned at a Li^+–O distance of 1.9 Å clearly indicates a strong affinity of a Li^+ ion around the carbonyl oxygen atom because of the preferred electrostatic interactions. The first peak location also increases from $g_{LiO}(r)$, $g_{LiC}(r)$, $g_{LiOS}(r)$, to $g_{LiC3}(r)$, while the magnitude decreases, suggesting that association of the Li^+ ion with these atoms becomes weaker. Integrating $g_{LiO}(r)$ to the first minimum reveals that there are about 4.2 EC molecules around a Li^+ ion in the first solvation shell. This observation is consistent with the Raman spectroscopy study by Hyodo and Okabayashi on $LiClO_4$ in EC.[29]

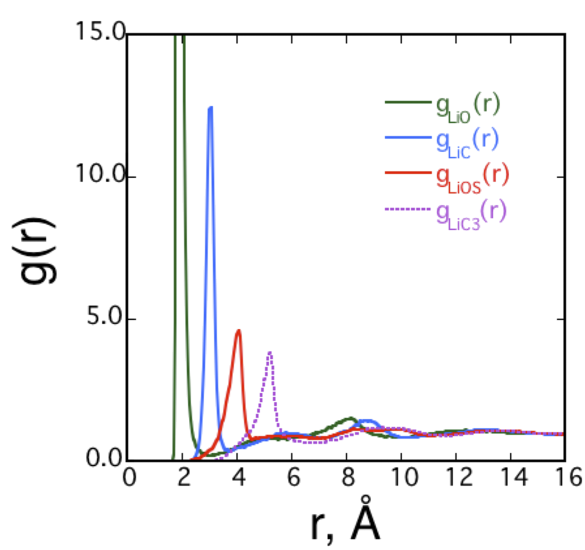

FIG. 4. Atomic radial distribution between Li^+ and EC.

C. Rate theory on Li^+ ions in liquid EC

To examine the EC solvent exchange dynamics around the Li^+ ion, we used Eq. (5) to calculate the ion-EC mean force as an average over the different solvent configurations,[30]

$$F(r) = \frac{1}{2}\left\langle \vec{r}_u \bullet (\vec{F}_A - \vec{F}_B)\right\rangle. \tag{5}$$

In this expression, F_A and F_B are the forces acting on the solutes A and B, respectively. The term $\vec{r}_u$, which is a unit vector along the A−B direction, is defined as

$$\vec{r}_u = \vec{r}_{AB}/\left|\vec{r}_A - \vec{r}_B\right|. \tag{6}$$

The PMF, $W(r)$, is calculated as

$$W(r) = -\int_{r_o}^{r_s} \langle F(r)\rangle dr. \tag{7}$$

We evaluated PMFs along the center-of-mass separation between the Li^+ ion and EC with the center-of-mass separation between them incremented by 0.1 Å. At each center-of-mass separation, the average $F(r)$ was determined from a 2-ns simulation time, preceded by a 500-ps equilibration period. The uncertainties on the PMFs were ±0.05 kcal/mol as estimated by determining the force averaged (the corresponding PMFs) over four equally spaced time frames during the production.

We further inspect the exchange dynamics of the EC molecules around the solvated Li^+ ion using several approaches including transition state theory, (TST), Grote-Hynes theory (GH), reactive flux, (RF), and Impey, Madden, and McDonald, (IMM) methods.[31–34] Figure 5 shows the computed PMF obtained at 330 K and at 1 bar. This PMF shows a free energy minimum at about 3.8 Å, with a transition state located at 4.3 Å and a barrier height for free energy of dissociation of 1.4 kcal/mol. For a given PMF, the rate constant for the exchange process can be computed using TST as follows:

$$\boldsymbol{k}^{TST} = \sqrt{\frac{\boldsymbol{k}_b T}{2\pi\mu}}\,\frac{(\boldsymbol{r}^*)^2 \boldsymbol{e}^{-\beta W(\boldsymbol{r}^*)}}{\int_0^{r^*} \boldsymbol{r}^2 \boldsymbol{e}^{-\beta W(r)} \boldsymbol{dr}}, \tag{8}$$

where r^* is defined as the position of the barrier top, μ is the ion-EC reduced mass, k_b is the Boltzmann constant, and T is the temperature. Using the computed PMFs and transition state distances, we computed the rate constant, k^{TST}, for the exchange process using Eq. (8); the result for k^{TST} is 0.9 ps^{-1}. The results are summarized in Table III.

TST is known to significantly overestimate the values of rate constants. The reason for this overestimation can be attributed to the basic assumption of TST that once the transition state is reached, the reactive species will proceed to form products without the possibility of re-crossing to the reactant side. Corrections to the TST, including the well-known RF method and GH theory, have been developed and reported in the literature. To correct the TST, these two approaches include a transmission coefficient, κ, that accounts for re-crossing at the transition state. In the RF method,[33] the transmission coefficient, κ_{RF}, is extracted from the plateau value of the time-dependent transmission coefficient as calculated using Eq. (9),

$$\kappa(t) = \frac{\langle \dot{r}(0)\theta[r(t) - r^*]\rangle_c}{\langle \dot{r}(0)\theta[\dot{r}(0)]\rangle_c}, \tag{9}$$

where $\theta(x)$ is the Heaviside step function, which is 1 if x is larger than 0 and 0 otherwise, and $\dot{r}(0)$ is the initial ion-EC velocity along the reaction coordinate. The subscript c means that the starting configurations have been generated in the constrained reaction coordinate ensemble. We constrained the reaction coordinate at its transition state value in the RF and GH approaches by removing the center-of-mass motion of the EC molecule that was pre-selected as well as the Li^+ ion to form a solute pair to perform the kinetics study in solutions. Initially, we generated a series of initial configurations from a constrained simulation in which the distance between a pre-selected EC molecule and the Li^+ ion is fixed at the transition state separation determined from the PMF calculation. This was achieved by running a 5-ns simulation, during which a configuration was collected every 2 ps to obtain 2500 configurations for computing the time-dependent transmission coefficient. Then, each of these 2500 initial configurations was simulated both backward and forward in time for 2 ps under the NVE ensemble, and the value of κ_{RF} was determined over the last 0.5 ps of the forward and backward trajectories. In Fig. 6(a), we present the computed time-dependent $\kappa(t)$. Note that the value of $\kappa(t)$ converges well, and the transmission coefficients κ_{RF}, estimated as described above, is 1.7×10^{-2} (see Table III).

Additionally, we also used GH theory to compute kinetic properties of the Li^+-EC exchange process. We evaluated the

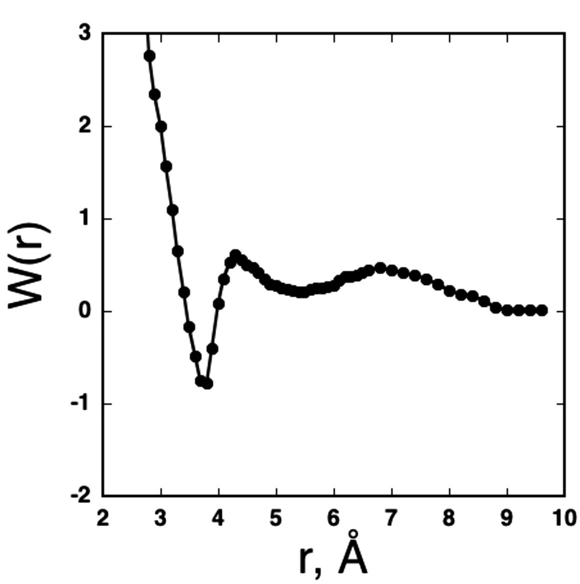

FIG. 5. Computed PMFs for Li^+-EC pair in EC at 330 K and 1 atm.

TABLE III. Computed rate coefficients at 330 K for Li^+ in EC.

	k^{TST}/ps	$\kappa_{RF*}k^{TST}$/ps	$\kappa_{GH*}\, k^{TST}$/ps	k^{IMM}/ps
k/ps	0.9	1.7×10^{-2}	1.5×10^{-2}	2.2×10^{-3}
Resident time (ps)	1	60	67	446

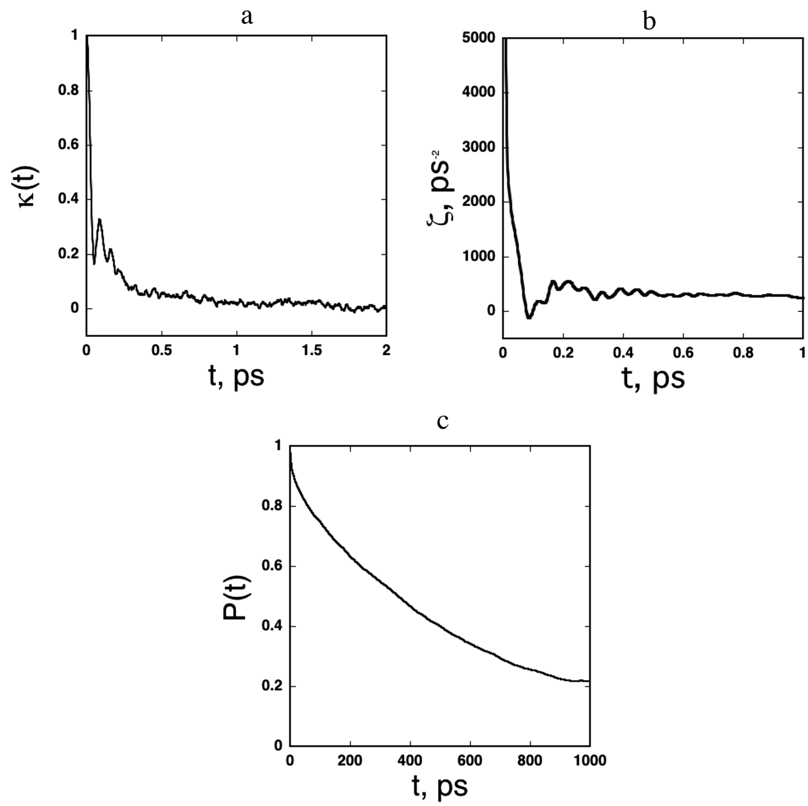

FIG. 6. (a) Computed time-dependent transmission coefficients, $\kappa(t)$, of Li$^+$-EC pair from the RF method. (b) Computed time-dependent friction kernels, $\xi(t)$, for Li$^+$-EC pair in EC computed from GH theory. (c) Computed time-correlation function as a function of pressure used to determine the residence time of an EC molecule is in the first solvation shell of the Li$^+$.

values for κ_{GH} and then compared the results with corresponding results obtained using the RF method. To obtain k_{GH}, we computed the friction kernel using the trajectories at the barrier region using Eqs. (10) and (11),

$$\zeta(t) = \frac{1}{\mu k_b T} \left\langle R(t, r^*) \cdot R(0, r^*) \right\rangle, \tag{10}$$

$$R(t, r) = F(t, r) - \langle F(t, r) \rangle. \tag{11}$$

The term r^* is the position of the barrier maximum, μ is the reduced mass, k_b is the Boltzmann constant, and T is the temperature. Here, $\xi(t)$ is the time-dependent friction coefficient and $R(t, r)$ is the fluctuating force. The GH theory transmission coefficient that considers re-crossings at the barrier top, k_{GH}, can be expressed as Eq. (12),[32]

$$\kappa_{GH} = \left(\kappa_{GH} + \int_0^\infty dt \frac{\zeta(t)}{\omega_b} e^{-\omega_b \kappa_{GH} t} \right)^{-1}. \tag{12}$$

The GH theory transmission coefficient involves the frequency component of the time-dependent friction coefficient, $\xi(t)$, at the Laplace frequency (reactive frequency), $\omega_b \kappa_{GH}$, relevant in the barrier region. The term ω_b is the barrier frequency obtained by fitting the PMF in the barrier region to an inverted parabola. The un-normalized, time-dependent friction kernels of Li$^+$-EC pair in liquid EC are presented in Fig. 6(b). Two distinct decay time scales are observed—an initial rapid decay lasting less than 0.1 ps and a longer time decay that lasts less than a picosecond. The oscillatory behavior of $\xi(t)$ reflects the barrier height magnitude of the computed PMF. From the invert parabola fit, we found the computed the barrier frequency, ω_b, to be 23.5 ps^{-1}. The computed transmission coefficient using GH theory is 1.6×10^{-2} and is presented in Table III. The rate constant results obtained using TST theory and then weighed with κ_{GH} also are shown in Table III. We found the corrected rate from GH theory to be very close to that of value obtained using the RF method.

In addition to the RF and GH methods, we also used the IMM method to investigate the kinetics.[33] The IMM method calculates the residence time of a solvent molecule in the first solvation shell of the solute (i.e., Li$^+$ ion). The rate constant then is calculated from the inversion of the residence time. The residence time can be determined from the normalized time-correlation function of the solvent molecule population in the first solvation shell of the Li$^+$, as given in Eq. (13),

$$P(t) = \left\langle n_\alpha(t_i, t_f, t^*) \right\rangle / \langle n_\alpha(t_i, t_i, t^*) \rangle_{\alpha, ti}. \tag{13}$$

The term n_α is the number of solvent molecules in the first solvation shell. It has a value of 1 if a solvent molecule is found in the first solvation shell at both the initial time, t_i, and the final time, t_f, and does not leave the shell for a continuous period of time longer than t^*. To ignore transient escapes from the first solvation shell, $t^* = 2$ ps is used. The function $\langle \ldots \rangle_{\alpha, ti}$ in Eq. (13) indicates averaging over solvent molecules (α) and initial times (t_i). $P(t)$ is approximated as the exponential decay function $\exp(-t/\tau_p)$, so the residence time τ_p can be determined using Eq. (14),

$$\tau_p = \int_0^\infty P(t) dt = \int_0^\infty \exp(-t/\tau_p) dt. \tag{14}$$

In this study, we used $t^* = 2.0$ ps (i.e., the value originally proposed by Impey *et al.*[34]).

The time-correlation function $P(t)$ is plotted as a function of time and is shown in Fig. 6(c). The decay is apparently quite slow, indicating a persistent first solvation shell. By integrating $P(t)$, we found that $\tau_p = 450$ ps. A comparison of the computed residence times for Li^+ ions in EC using the RF, GH, and IMM methods are summarized in Table III. The observed trends are very consistent; it is clear that re-crossing significantly reduces the exchange rate coefficient. By ignoring re-crossing at the barrier, TST significantly overestimates the rate in the Li^+-EC system. RF and GH methods improve the rates by considering the re-crossing and predict much longer residence times (smaller rates). IMM, on the other hand, directly computes and extracts the residence time from the time correlation function. It is found that the numerical value obtained using the IMM method is consistently higher, which may be due to the strong association of Li^+ and EC.

D. Kinetics of ion pairing in liquid EC

We begin this section by presenting results for the kinetics of the Li^+–$[BF_4]$/$[PF_6]$ ion pairs in liquid EC. We start with the Li^+–$[BF_4]$/$[PF_6]$ PMFs, and then continue with rate theory results determined using the RF and GH methods. We end this section with a comparison of results obtained using the different rate theory methods. In Figs. 7 and 8, we present PMF plots as a function of center-of-mass separation between the two ions in liquid EC. We found that the PMF for the ion pair containing $[BF_4]$ as the anion has a deeper contact ion pair minimum compared to the corresponding $[PF_6]$-based LIB. This demonstrates that the anion with a small (spherical) shape is more strongly associated with the Li^+ in EC. The free energy barriers for the ion pair dissociation are found to be about 3.4 and 1.7 kcal/mol for Li^+–$[BF_4]$ and Li^+–$[PF_6]$, respectively. Correspondingly, the computed rate constants from the PMFs using TST are determined to be 5.4×10^{-2} and 4.6×10^{-1} ps^{-1}. These results clearly indicate that a free energy difference of 1.7 kcal/mol can have a noticeable effect on the rate constants. In addition, we noted that no well-defined solvent-separated ion pair was present as routinely observed for ion pairs in non-aqueous solutions. This is probably due to the absence of strong

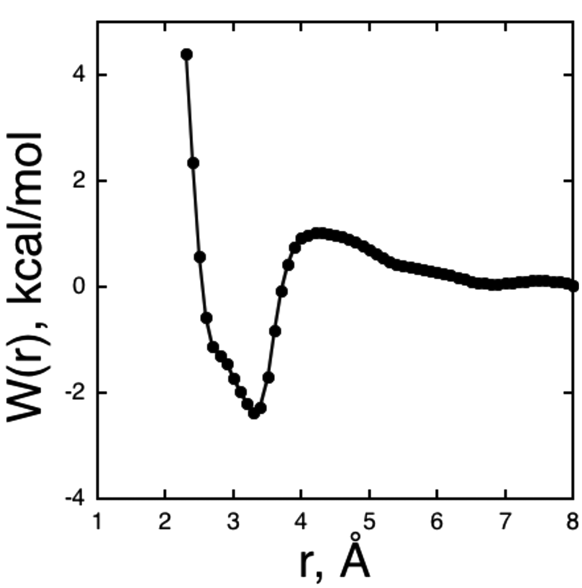

FIG. 7. Computed PMFs for Li^+–$[BF_4]$ in EC at 330 K.

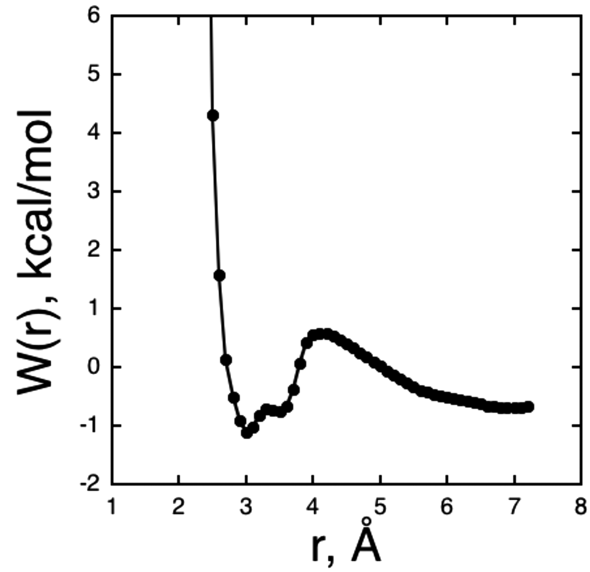

FIG. 8. Computed PMFs for Li^+–$[PF_6]$ in EC at 330 K.

preferred interactions between ion-solvent and solvent-solvent systems.

As mentioned above, we also computed the rate coefficients of the ion pair using the RF and GH theories. In Figs. 9(a) and 9(b), we present the computed time-dependent $\kappa(t)$ for both ion pairs using the RF method. Subtle differences are observed in these transmission coefficient functions. The transmission coefficients, κ_{RF}, are estimated to be 8.4 $\times$ 10^{-2} and 2.5 $\times$ 10^{-2} for the Li^+–$[BF_4]$ and Li^+–$[PF_6]$ ion pairs, respectively. These results are consistent with the computed PMFs, and we observed that the rate constants increase from Li^+–$[BF_4]$ to Li^+–$[PF_6]$. Our results show that the anion type, in addition to affecting the free energy of ion pair dissociation in EC, also significantly influence the kinetics of ion pairings. The persistence of the ion pair can significantly affect the dynamical properties of the ions and thus affect the ionic conductivity as observed in previous studies. In Table IV, we present the rate constant results obtained using RF theory for the Li^+–$[BF_4]$ and Li^+–$[PF_6]$ ion pairs. These results demonstrate that the solvent response to the kinetics of ion pairing is significant.

In Figs. 9(c) and 9(d), we show plots of un-normalized, time-dependent friction kernels of Li^+–$[BF_4]$ and Li^+–$[PF_6]$ ion pairs in EC for comparison. In all cases, two distinct decay time scales are observed with an initial fast decay of a time scale <0.1 ps, followed by a slower decay that lasts for a few picoseconds. However, there are distinct features between these two friction kernels. For the Li^+–$[BF_4]$ ion pair, a deeper well is observed in $\xi(t)$ at about 0.1 ps. We also observed that the barrier frequency, ω_b, increases from 10.5 ps^{-1} to 15.6 ps^{-1} as the system changes from Li^+–$[BF_4]$ ion pairs to Li^+–$[PF_6]$ ion pairs. The computed transmission coefficients using GH theory are provided in Table IV. The computed values for κ_{GH} are 1.1 $\times$ 10^{-2} and 4.1 $\times$ 10^{-2} for Li^+–$[BF_4]$ and Li^+–$[PF_6]$, respectively. In Table IV, we present the rate constant results obtained using TST theory and then corrected with κ_{GH}. We found that the relaxation times changed from 1700 ps to 53 ps for Li^+–$[BF_4]$ and Li^+–$[PF_6]$ ion pairs,

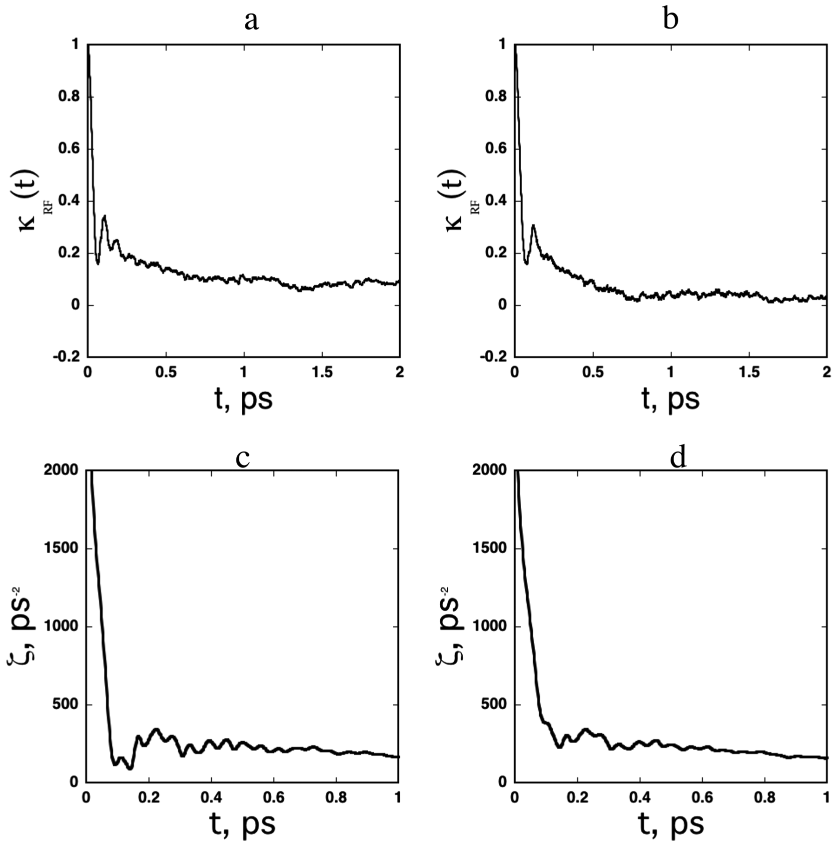

FIG. 9. (a) Computed time-dependent transmission coefficients at 330 K, $k(t)$, for Li^+–$[BF_4]$ and (b) for Li^+–$[PF_6]$ in EC computed from the RF method. (c) Time-dependent friction kernels, $\xi(t)$, for Li^+–$[BF_4]$ and (d) for Li^+–$[PF_6]$ ion pairs in EC computed from GH theory.

respectively. Both RF and GH theories predict transmission coefficients that are quite smaller than 1, indicating a high probability of re-crossing at the transition state. Additionally, the GH theory results show the same trend as the transmission coefficients computed using the RF method, which is that the rate constant increases from Li^+–$[BF_4]$ to Li^+–$[PF_6]$ ion pairs. Interestingly, a similar behavior has also been observed for the Li^+–$[BF_4]$ and Li^+–$[PF_6]$ pairs in acetonitrile solutions, which may be attributed to the ionic interaction. This may also imply that the anion can play a role in the Li^+-ion mobility in solutions.

In this study, we also find that although TST tends to overestimate the reaction rate due to the neglect of re-crossing at the barrier, it is sufficient to establish a general trend for reaction rates in these systems. By considering the re-crossing events, RF and GH methods predict much smaller rates, indicating the presence of significant re-crossing. The IMM approach directly computes and extracts the rate from the time correlation function and predicts the reaction rate that is smaller than both RF and GH methods.

TABLE IV. Computed rate coefficients at 330 K for Li^+–$[BF_4]$ and Li^+–$[PF_6]$ ion pairs in EC.

Ion-pair	k^{TST}/ps	κ_{RF}	$\kappa_{RF*}k^{TST}$/ps	κ_{GH}	$\kappa_{GH*}\ k^{TST}$/ps
Li^+–$[BF_4]$	5.4×10^{-2}	8.4×10^{-2}	4.5×10^{-3}	1.1×10^{-2}	6.0×10^{-4}
Relax (ps)	19		220		1700
Li^+–$[PF_6]$	4.6×10^{-1}	2.5×10^{-2}	1.1×10^{-2}	4.1×10^{-2}	1.9×10^{-2}
Relax (ps)	2.1		87		53

IV. CONCLUSIONS

In summary, we carried out systematic studies using MD simulations and rate theory methods to investigate the mechanism of EC exchange around Li^+ ions and the kinetics of Li^+ $[BF_4]$ and Li^+ $[PF_6]$ systems. We started with the construction of a polarizable EC force field that is shown to reproduce many experimental thermodynamic and structural properties of EC. To obtain a thorough perspective of the EC exchange dynamics around the Li^+ ion, we computed many properties associated with the EC exchange process, including ion-EC PMFs, time-dependent transmission coefficients, and rate constants. We use TST, RF, and GH methods to estimate the rate constants and find that the correction to TST can be significant in these systems. We also examined the Li^+–$[BF_4]$ and Li^+–$[PF_6]$ ion-pairing kinetics in liquid EC. We found that the relaxation times varied from 1700 ps to 53 ps (based on GH theory) and from 220 ps to 87 ps (based on the RF approach) for the Li^+–$[BF_4]$ and the Li^+–$[PF_6]$ ion pairs, respectively. These results confirm that the solvent response to the kinetics of ion pairing is significant. The GH theory results qualitatively agree with the transmission coefficients obtained using the RF method. We also observed that both theories present a considerable correction to the TST. Our results also show that anion type, in addition to affecting the free energy of dissociation in EC, also significantly influences the kinetics of ion pairings.

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